

Documentation

Investigation 1: Customer Loyalty & Repeat Customer Analysis

Business Problem

The Marketing team wanted to understand customer loyalty by measuring the repeat customer rate and identifying the top repeat customers.

Business Questions

- What is the repeat customer rate?
- Who are our top repeat customers?
- How loyal are our customers?

SQL Concepts Used

- Joins
- CTEs
- Aggregations
- HAVING
- Subqueries

Key Findings

- Repeat Customer Rate: **3.12%**
- A very small percentage of customers placed multiple orders.
- Most customers made only one purchase.

Business Recommendations

- Launch customer retention campaigns.

- Offer personalized discounts to encourage second purchases.
 - Develop a loyalty program targeting repeat customers.
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Investigation 2: Monthly Sales Performance

Business Problem

The Sales team wanted to identify the best-performing months based on revenue and delivered orders.

Business Questions

- Which month generated the highest revenue?
- Which month had the highest number of delivered orders?
- Are there seasonal trends?

SQL Concepts Used

- GROUP BY
- Date Functions
- Aggregations

Key Findings

- August 2018 generated the highest revenue and delivered orders.
- Significant revenue decline occurred in September.

Business Recommendations

- Investigate promotional campaigns and seasonality.
 - Replicate successful marketing strategies from high-performing months.
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Investigation 3: Delivery Performance by State

Business Problem

Operations wanted to understand delivery performance across different states.

Business Questions

- Which states experience the longest delivery times?
- Where are delivery delays occurring?

SQL Concepts Used

- `TIMESTAMPDIFF()`
- `GROUP BY`
- Aggregations

Key Findings

- Delivery delays were highest in AM, PA, and PB.
- Most delays occurred after carrier pickup.

Business Recommendations

- Evaluate carrier performance.
 - Improve logistics in underperforming regions.
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Investigation 4: Monthly Revenue Trend

Business Problem

Management wanted to analyze month-over-month revenue performance.

Business Questions

- How is revenue changing over time?
- Which months experienced significant growth or decline?

SQL Concepts Used

- CTEs
- LAG()
- Window Functions
- Date Functions

Key Findings

- August showed strong revenue growth.
- September experienced a significant decline.

Business Recommendations

- Investigate demand fluctuations.
 - Compare promotional activities across months.
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Investigation 5: Product Category Performance

Business Problem

The Product team wanted to identify the strongest and weakest product categories.

Business Questions

- Which categories generate the most revenue?
- Which categories receive the most orders?
- Which categories underperform?

SQL Concepts Used

- Joins
- Aggregations
- DENSE_RANK()

- Window Functions

Key Findings

- Beauty, Bed & Bath, Sports, Watches, and Computers were top-performing categories.
- CDs/DVDs and several niche categories generated minimal revenue.

Business Recommendations

- Invest more in high-performing categories.
 - Review low-performing categories before discontinuation.
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Investigation 6: Product Category Trend Analysis

Business Problem

The Product Director wanted to understand how product categories performed over time.

Business Questions

- Which categories are consistently growing?
- Which categories are declining?
- Are there any unusual revenue trends?

SQL Concepts Used

- LAG()
- Window Functions
- CASE
- CTEs

Key Findings

- Several categories experienced sustained monthly declines.
- Some categories showed strong month-over-month growth.

Business Recommendations

- Investigate declining categories.
 - Increase inventory for consistently growing categories.
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Investigation 7: High-Value Customer Analysis

Business Problem

The CRM team wanted to identify customers eligible for a loyalty program.

Business Questions

- Who generates the highest revenue?
- Are valuable customers frequent buyers or high spenders?

SQL Concepts Used

- CTEs
- DENSE_RANK()
- RANK()
- Aggregations

Key Findings

- Top customers generated high revenue through a small number of purchases.
- High spending and purchase frequency were not always correlated.

Business Recommendations

- Reward high-value customers.

- Create customer segments based on revenue and purchase frequency.
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Investigation 8: Product Growth & Decline Analysis

Business Problem

The merchandising team wanted to monitor product category performance over time.

Business Questions

- Which categories are growing?
- Which categories are declining?
- Which require immediate attention?

SQL Concepts Used

- LAG()
- CASE
- Window Functions
- CTEs

Key Findings

- Some categories experienced consistent monthly declines.
- Several categories demonstrated strong positive growth.

Business Recommendations

- Investigate declining categories.
 - Align inventory planning with demand trends.
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Investigation 9: Logistics & Delivery Delay Investigation

Business Problem

The Operations team wanted to identify bottlenecks in the delivery process.

Business Questions

- Where are delays occurring?
- Which states are most affected?
- What operational improvements are needed?

SQL Concepts Used

- `TIMESTAMPDIFF()`
- Aggregations
- `DENSE_RANK()`
- CTEs

Key Findings

- The largest delays occurred between carrier pickup and customer delivery.
- Certain states consistently experienced longer delivery times.

Business Recommendations

- Review carrier performance.
 - Investigate regional logistics challenges.
 - Analyze delays by product category for deeper insights.
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Investigation 10: Payment Behavior Analysis

Business Problem

The Finance team wanted to understand customer payment preferences and identify potential financial risks.

Business Questions

- Which payment methods are most popular?
- Which contribute the most revenue?
- How do customers differ in payment behavior?

SQL Concepts Used

- Aggregations
- CTEs
- Window Functions
- Revenue Contribution Analysis

Key Findings

- Credit cards accounted for approximately **78%** of total revenue.
- Debit cards had the lowest adoption.
- Voucher payments generated more revenue than debit cards, while debit card purchases had higher average order values.

Business Recommendations

- Continue optimizing the credit card payment experience.
- Investigate the low adoption of debit cards.
- Evaluate voucher programs to ensure they drive incremental revenue.